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Gap capture in northern hardwoods: patterns of establishment and height growth in four species

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Abstract

We sought to determine whether four tree species that commonly coexist—American beech (*Fagus grandifolia* Ehrh.), sugar maple (*Acer saccharum* Marsh.), red maple (*Acer rubrum* L.), and yellow birch (*Betula alleghaniensis* Britt.)—differed in their patterns of establishment in gaps and their height growth after gap formation. Fifty-eight canopy trees were sampled in four experimentally created canopy gaps, 44–48 years old, in a northern hardwood forest at the Bartlett Experimental Forest, NH. Using stem analysis, we determined the year of origin of each tree and its pattern of height growth. Over two-thirds of beech and sugar maple canopy trees were advance regeneration, while most yellow birch stems established after gap formation. Red maple was a mixture of the two. Beech had the oldest stems at the time of gap formation (5 out of 18 stems over 20 years of age). No canopy tree established later than four years after gap formation. Species differed in their patterns of height growth, with beech growing more slowly than the others immediately after gap formation. Beech, however, grew at a more constant rate than sugar maple and yellow birch, and approached the heights of these species 30 years after gap formation. Red maple was the tallest and maintained a constant growth rate over the 30-year period. In beech and sugar maple, individuals that established after gap formation eventually grew faster and caught up with those establishing via advance regeneration. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: Canopy gaps; Advance regeneration; Height growth; *Fagus grandifolia*; *Acer saccharum*; *Acer rubrum*; *Betula alleghaniensis*

1. Introduction

Periodic disturbances in forest communities may damage or kill trees, creating canopy gaps. While small gaps may be filled by horizontal growth of

the crowns of adjacent trees, larger gaps are filled primarily by the upward growth of small trees. Trees that ‘capture’ a large gap—those whose crowns eventually occupy space in the forest canopy—may do so in two ways. Advance regeneration, seedlings and saplings that have grown slowly in the shady understory, may grow rapidly upon canopy opening. Alternatively, seedlings may establish at or after the time of gap formation and grow rapidly enough to catch and keep up with advance regeneration.

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The importance of advance regeneration in gap colonization has been studied in several different forests. Runkle (1989), working in eastern US deciduous forests, and Nakashizuka (1983), working in Japanese beech forests, found advance regeneration had an advantage in filling gaps. In an Amazonian forest, advance regeneration accounted for 97% of all trees in small (single tree-fall) gaps and 83% in a larger (multiple tree-fall) gap (Uhl et al., 1988). Rates of height growth among species establishing by advance regeneration and post-gap colonization, however, have not been as well documented.

The goal of our research was to determine whether four species that commonly coexist in late successional northern hardwood forests in New Hampshire—American beech (*Fagus grandifolia* Ehrh.), sugar maple (*Acer saccharum* Marsh.), red maple (*Acer rubrum* L.), and yellow birch (*Betula alleghaniensis* Britt.)—differed in their patterns of gap capture.

American beech reproduces both by seed and vegetatively through root sprouts or suckers (Burns and Honkala, 1990). Seedlings and saplings are extremely shade-tolerant and can survive in shade for over 100 years (Poulson and Platt, 1989). Understory beech show increased growth when light levels increase (Canham, 1988).

Sugar maple reproduces by seed and its seedlings survive long periods of slow growth (Burns and Honkala, 1990; Forcier, 1975). It is only slightly less tolerant of suppression than beech, growing more slowly in shade, but growing twice as fast in gaps (Canham, 1985; Canham, 1989).

Red maple reproduces by seed and is also a vigorous stump sprouter (Burns and Honkala, 1990). Its seedlings can survive in shade for several years, but it is only moderately tolerant when compared to beech and sugar maple (Burns and Honkala, 1990). Red maple has rapid early growth and responds to release with increased growth after the pole stage (Burns and Honkala, 1990).

Yellow birch seedlings require complete release from overstory shading to survive and grow (Burns and Honkala, 1990). This species is intermediate in shade tolerance and thus cannot establish in an undisturbed forest (Forcier, 1975). Six years after clearcutting, however, yellow birch attained greater heights than sugar maple and beech (Bormann and Likens,

1994). This tendency to outgrow sugar maple and beech is stronger on moist sites than on dry sites (Hill, 1987).

While much is known about these species' requirements for establishment, growth, and juvenile survival, little data exist on the relative importance of advance regeneration in successful gap capture (Connell, 1989; White et al., 1985). Clearly, all four species can respond to resources liberated by the formation of canopy gaps, either through release of advance regeneration or through establishment of new seedlings. The relative importance of the two modes of gap capture, however, is not well known.

Although there are few studies on establishment patterns, tree growth, and success in canopy gaps, much is known about short-term responses to clearcutting. Bormann and Likens (1994) showed that, at six years after clearcutting, 86% of the tallest beech saplings were from advance regeneration, while the majority of sugar maple and yellow birch stems were of seed origin. This pattern may not persist to tree maturity, however, as the advance regeneration and new germinants of any one species may have different rates of survival after Year 6. Moreover, what is true for large openings such as clearcuts is not necessarily the case in smaller canopy openings.

In 45–49-year-old gaps (Table 1) created by group selection cuttings in the White Mountains of New Hampshire, we found individuals of American beech, yellow birch, and sugar and red maple that were dominant or codominant in the forest canopy (Leak, 1991; McClure and Lee, 1993). Using stem analysis of these successful individuals, we tested the hypotheses that the four species differed in their (1) times of establishment relative to time of gap formation, and (2) patterns of height growth over the 45–49 years since gap formation.

2. Methods

2.1. Site description

Research was conducted in a 46-ha tract in the Bartlett Experimental Forest, Bartlett, NH, operated by the U.S. Forest Service, Northeastern Forest Experiment Station. We studied four canopy gaps created by group selection cuttings during the years

Table 1

Characteristics of four anthropogenic gaps studied at the Bartlett Experimental Forest, Bartlett, NH

Gap	G	2	6	7
Substrate type	coarse washed till/loose rock	coarse washed till	enriched till	coarse washed / enriched till
Slope angle (%)	30	14	15	14
Aspect	NNE	WNW	NNW	W
Years since gap formation	45	48	46	49
Year of gap formation	1941	1938	1940	1937
Gap area (m ²)	1619	1781	2428	1862
Species (# of stems)				
American beech	6	4	2	6
Sugar maple	2	3	7	0
Red maple	0	4	5	5
Yellow birch	4	3	3	4

1937–1941 (Jensen, 1943). Gaps were chosen for their similarity of age, size, and environmental characteristics (Table 1, and McClure and Lee, 1993). Although it would have been preferable to include more gaps in our study, other gaps of similar age and size did not exist at this site. The gaps ranged in size from 1619 to 2428 m², and in slope from 14% to 30% and had a west, north, or northeast aspect. Gap sizes were determined by Forest Service maps of the original cuts. Absence of subsequent windthrow at gap edges was verified by comparison of gap shape in the field, as indicated by large margin trees, with shapes shown on the maps. Soils were sandy loams, derived from glacial drift composed of feldspar-rich granite (Filip and Little, 1971). Gaps occurred on enriched till, coarse washed till, or loose rock habitats (Leak, 1982; McClure and Lee, 1993).

2.2. Vegetation sampling

Fifty-eight trees in the four gaps were destructively sampled in 1985. All of the tallest, dominant or co-dominant trees of each species were sampled. Trees were not sampled if they appeared to be butt or stump sprouts from trees cut when the gap was created. We harvested a total of 18 beech, 12 sugar maple, 14 red maple, and 14 yellow birch trees. There were 2–7 individuals of each species in each gap with the exception of sugar maple and red maple, which were not present as dominants or codominants in two different gaps (Table 1). Trees were cut with a chain-

saw as close to the base of the tree as possible. A basal cross section was cut as well as sequential sections at 2-m intervals, beginning at 1.5 m height. Despite our efforts to cut each tree as close to the base of the bole as possible, basal sections may have missed one to a few years of the earliest growth.

Sections were planed and sanded to a smooth finish. Rings were counted under a dissecting microscope on two radii >45° apart.

2.3. Data analysis

Tree ages at the time of gap formation were determined by ring counts and compared using analysis of variance (SYSTAT 5.2, Wilkinson et al., 1992). Tree ages were square-root transformed to equalize variances. The relationship between height growth and number of years since gap formation was analyzed. Given our objective, it would seem that tree height would have been our dependent variable. However, as tree age at any particular height could be determined only after trees were cut, we sampled the trees at fixed heights and counted rings at these heights. Thus, the dependent variable in our analysis was the number of years (rings produced) before or since gap formation. The number of years from gap formation at each of the fixed heights was regressed against species, gap, age at the time of gap formation, and related interactions using repeated measures regression (SYSTAT 5.2, Wilkinson et al., 1992). Repeated measures regression was used since ring number was measured at each of

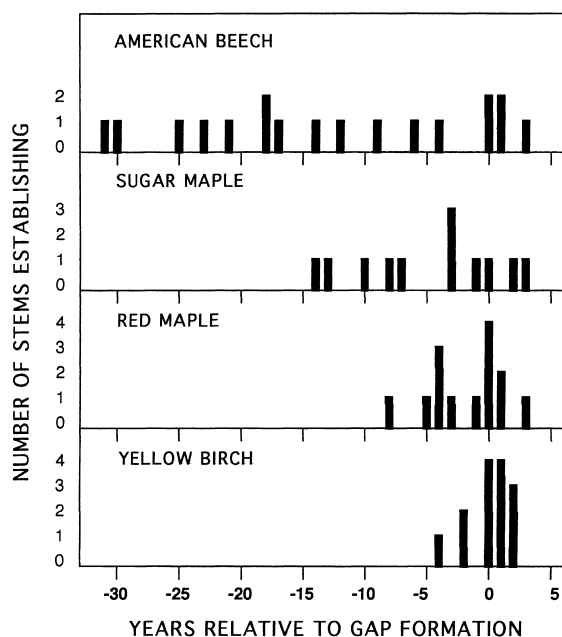


Fig. 1. Number of individual trees of American beech, sugar maple, red maple, and yellow birch establishing at different times relative to gap formation. Data collected at the Bartlett Experimental Forest, Bartlett, NH.

the fixed heights on each tree. As trees varied in height, the analysis was run using different maximum heights. We used a maximum height of 15.5 m as it allowed examination of a broad range of tree heights with minimal reduction in sample size. Three beech trees and one yellow birch tree were dropped from this analysis because their heights were <15.5 m. They are included, however, in our reporting of general trends in Table 1 and Fig. 1.

3. Results

3.1. Time of establishment

Stem age at the time of gap formation varied significantly between species (ANOVA, $p < 0.0001$, $R^2 = 0.46$, Fig. 1). On average, beech were the oldest (mean = 11.1 years) at the time of gap formation, yellow birch the youngest (mean = 0.2 years), and sugar and red maple were intermediate (mean = 2.3, 1.4 years, respectively; all means adjusted for effects

of gap). The number of stems established prior to gap formation also differed among species. Seventy-five percent of sugar maple stems, 72% of beech stems, 50% of red maple stems and only 21% of yellow birch stems were advance regeneration.

In addition, the age ranges of the advance regeneration (those stems established prior to gap formation) differed among species (Fig. 1). Nearly half of beech advance regeneration established 15 years or more before gap formation; several trees were over 20 years old and the oldest was 31 years old at the time of gap formation. The oldest sugar maple advance regeneration established 13 years before gap formation, with 25% established 10–13 years before, 22% established 5–9 years before, and 44% established 1–4 years before. The oldest red maple advance regeneration established eight years before gap formation with most having established 1–4 years prior to gap formation. Yellow birch advance regeneration contrasted most strongly with beech, with establishment occurring <5 years prior to gap formation. Most yellow birch established after gap formation. In all species, the individuals established after gap formation did so within four years.

Gap, as a covariate, was also significant ($p = 0.017$). Regardless of species, ages of individuals at the time of gap formation differed among gaps (Fig. 2). Stem age at the time of gap formation, adjusted for differences among species, was greatest in Gap 6, lowest in Gap 2 and intermediate in gaps G and 7.

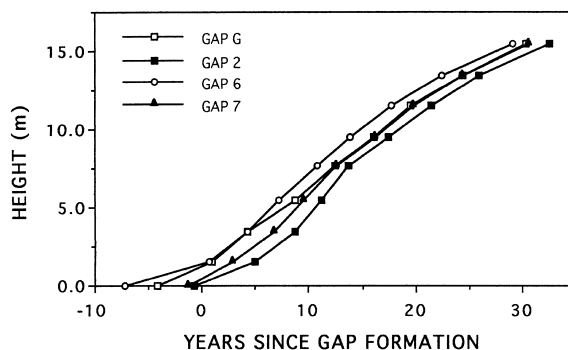


Fig. 2. Mean tree height as a function of years since gap formation for four gaps (G, 2, 6 and 7) at the Bartlett Experimental Forest, Bartlett, NH. Values are means adjusted for the effects of species.

Table 2
Repeated measures analysis of tree age (number of annual rings)

Source	SS	df	MS	F	P
<i>Between subjects</i>					
Gap	741.8	3	247.3	4.24	0.010
Species	633.9	3	211.3	3.63	0.020
Tree age at time of gap formation	107.0	1	107.0	1.84	0.182
Error	2679.5	46	58.2		
<i>Within subjects</i>					
Height	19 628.1	7	2804.0	796.57	0.000
Height*gap	61.7	21	2.9	0.83	0.676
Height*species	132.4	21	6.3	1.79	0.019
Height*tree age at gap formation	126.6	7	18.1	5.14	0.000
Error	1133.5	322	3.520		

Note: Ages of individual trees (subjects) were measured at different heights (1.5–15.5 m). Independent variables were height, gap, species, and tree age at the time of gap formation. Tree species were American beech, sugar maple, red maple, and yellow birch. Gaps were 2, 6, 7, and G. Data collected at the Bartlett Experimental Forest, Bartlett, NH. $n = 54$, $R^2 = 0.477$.

3.2. Height growth

The relationship between height and years since gap formation varied significantly among species (Table 2). A height*species interaction ($p < 0.019$) indicated that each species had a different pattern of growth since the time of gap formation (Fig. 3). Thus, the shapes of the species' growth curves differed significantly. On average, red maple maintained the greatest height and had a constant growth rate for the entire period after gap formation, reaching 15.5 m in 26 years. Sugar maple tree heights, however, were

similar to those of red maple until 24 years after gap formation when sugar maple growth slowed; 30 years after gap formation it reached 15.5 m. Yellow birch heights were lower than the other three species in the decade following gap formation. It grew more rapidly than sugar and red maple between 4 and 12 years, nearly catching up to them in Year 12, and then maintained a growth rate similar to sugar and red maple for another 10 years before slowing down. Yellow birch had greater heights and showed more rapid growth than beech between years 5 and 12, but it grew more slowly than beech for the remaining years. Beech had the lowest average height most of the time, but like red maple had a constant growth rate. Thirty-three years after gap formation, beech attained a height of 15.5 m, was growing more rapidly than yellow birch, and appeared poised to overtop it soon.

An interaction between height and tree age at the time of gap formation (Table 2, $p < 0.0001$) indicated that the relationship between tree height and number of years from gap formation varied significantly depending on time of tree establishment (Fig. 4). In general, stems with a later time of establishment grew faster, often catching up with advance regeneration. Beech and sugar maple had the oldest advance regeneration of the four species. Beech advance regeneration, once released, grew quickly, remaining taller than beech individuals established at the time of

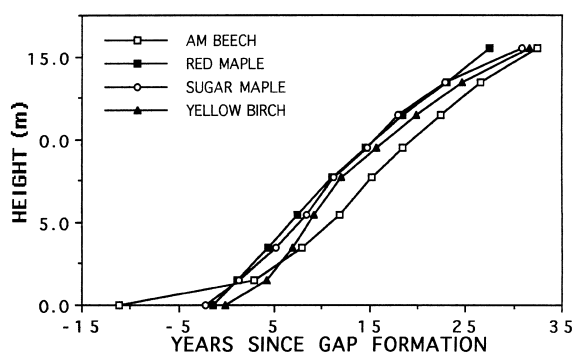


Fig. 3. Mean tree height as a function of years since gap formation for American beech, sugar maple, red maple, and yellow birch at the Bartlett Experimental Forest, Bartlett, NH. Values are means adjusted for the effects of gap.

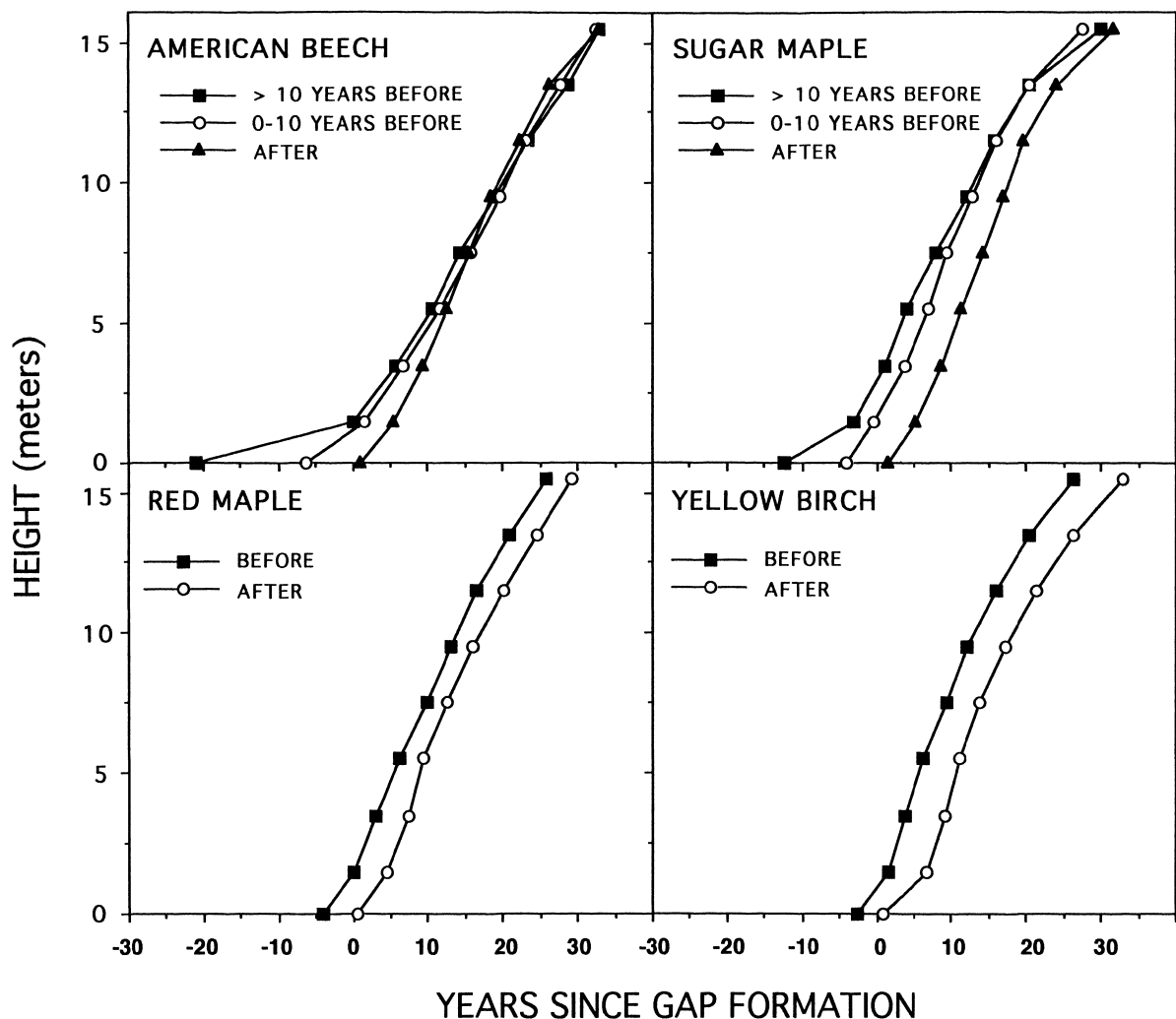


Fig. 4. Mean tree height as a function of years since gap formation and time of establishment for American beech, sugar maple, red maple, and yellow birch at the Bartlett Experimental Forest, Bartlett, NH. Values are means for groups of individuals establishing at different times. 'Before' refers to stems establishing before gap formation. 'After' refers to stems establishing after gap formation.

gap formation for about 15 years. Beech established at the time of gap formation grew more rapidly, however, approaching the heights of advance regeneration at that time. Sugar maple advance regeneration remained taller than individuals establishing after gap formation for the entire time under study. The difference between the two was eliminated for the most part after 30 years. Red maple and yellow birch advance regeneration, only a year or two older than stems originating at the time of gap formation, remained taller throughout the study period.

The significance of gap in the repeated measures analysis indicated that trees in different gaps differed in the time from gap formation they required to reach particular heights. Even though, at any given time after gap formation, trees were taller in some gaps than others, growth rates of trees were not different among gaps. This is shown by the absence of a height*gap interaction in this analysis (Table 2). Differences in tree height among gaps seemed to be due more to differences in the time of tree establishment than to differences in growth rate. Regardless of species,

individuals growing in different gaps had growth curves of similar slope but differed in the time they required to reach particular heights (Fig. 2). Generally, those individuals with earlier times of establishment maintained their initial advantage. Trees in Gap 6 were established earlier than those in gaps 7 or 2 and this initial height advantage was maintained.

4. Discussion

The four species in dominant or codominant positions in the canopy of these former gaps differed in their times of establishment and in their growth patterns after gap formation. Height growth patterns were further influenced by each individual's time of establishment.

Each species had a different range of times of establishment relative to gap formation. American beech survived in the understory for up to 30 years and, once released, became dominant in the canopy. Thus, dominant beech individuals in the canopy may be up to 30 years older than their counterparts in other species. Sugar maple also relied largely on advance regeneration to achieve dominance in the canopy, although none of our individuals had survived >15 years in the shade prior to gap formation. Half of the dominant red maple was established prior to gap formation, but the mean age at the time of gap formation was only 4.1 years. Red maple is known to benefit from a period of growth prior to release (Smith and Ashton, 1993) and to respond positively to release (Oliver and Stephens, 1977; Lorimer, 1980). However, this proclivity is equally balanced in our study by its ability to grow quickly from seed. Of the four species, yellow birch appeared to be most dependent on disturbance for establishment. Only 21% of dominant yellow birch stems were established before gap formation and, at the time of gap formation, the mean age of advance regeneration was the lowest of the four species, at 2.6 years.

Almost half of all sampled trees were established after gap formation, and within four years. Thus, no dominant tree was more than four years younger than the gap itself. Smaller gaps would likely decrease the percentage of dominant individuals established after gap formation (Forcier, 1975; Oliver and Stephens, 1977).

Height-growth patterns after gap formation were also different for each species. Red maple and yellow birch, which for the most part established around the time of gap formation, grew quickly between 5 and 15 years after gap formation, the latter species growing more rapidly. While the growth rate of yellow birch slowed slightly after 12 years, that for red maple continued at a high level (Fig. 3). This high rate of growth, however, would probably not be sustained. Palik and Pregitzer (1992), for example, observed decreasing height increment growth of red maple after 30 years in aspen forests of northern Michigan.

After gap formation, sugar maple grew more rapidly than beech even though beech included relatively more and older advance regeneration (Fig. 3) and was, thus, probably taller than sugar maple at the time of gap formation. Sugar maple's growth rate was most similar to that of red maple until ≈ 23 years after gap formation when its growth rate slowed.

We found that mean yellow birch height exceeded that of beech for most of the period after gap formation, but never exceeded that of sugar or red maple. In contrast, Hill (1987), who also worked at the Bartlett Experimental Forest and included in her sample some of the gaps we used, found that yellow birch height exceeded that of both beech and sugar maple throughout the 35-year period after gap formation.

Sugar maple and beech height-growth patterns were distinguished from those of yellow birch and red maple by the ability of the individuals established after gap formation to catch up to the height of the well-established advance regeneration (Fig. 4). In <20 years, beech individuals that established after gap formation caught up with advance regeneration. Sugar maple established after gap formation took longer, but at 30 years the later established individuals had nearly reached the height of the advance regeneration. These patterns suggest that beech and sugar maple individuals, establishing in deep shade and gaining an initial height advantage, may not grow as rapidly as some individuals establishing near the time of gap formation.

Trees reached a given height sooner in some gaps than in others, but the similar growth rates (parallel growth curves) in all gaps suggested that environmental conditions did not vary sufficiently among gaps to cause differences in tree growth rates after gap formation. We expected to see a steeper growth curve in

Table 3

Site indices (site index using 50-year b.h. as reference age) for American beech, sugar maple, red maple, and yellow birch, following Carmean (1978) and Leak (1978)

Species	Site index estimates	
	Carmean (1978)	Leak (1978)
American beech	70	65 on coarse washed till
Sugar maple	70	50 on coarse washed till
		72 on enriched till
Red maple	80	—
Yellow birch	60	55 on coarse washed till
		72 on enriched till

Gap 6 (enriched habitat) and a less steep growth curve in Gap G (loose rock and coarse washed till habitats), but growth rates were not consistent with these expectations (Fig. 2). Instead, the relationship between tree height and years since gap formation was associated with time of tree establishment. Gaps with earlier tree establishment tended to have taller trees. Why some gaps had older advance regeneration than others is not known, but it is interesting that Gap 6, which had the oldest advance regeneration, also had the most favorable soil conditions for growth.

Site index curves plot the measured height of a tree against its age. The greater the height at a given age, the more productive a site. We compared height–age curves of our species to standard site index curves in order to determine their accuracy and the site’s productivity. Site index is typically calculated as the height in feet of the average tree at age 50 (Erdman and Peterson, 1982). Thus, extrapolation of our curves was necessary for comparison. The Carmean (1978) site index curves were similar to ours for yellow birch and sugar and red maple. These species seem to have more rapid early height growth and slower later height growth (Carmean, 1979). Our yellow birch height-growth curve, however, slowed down at an earlier age than Carmean’s. Beech, in our study, sustained more rapid height growth in the early years than the trees studied by Carmean (1978). If it sustained this growth in a linear fashion, as Carmean (1979) would predict, our beech would have a projected site index of 70, slightly higher than Carmean’s, but consistent with the Solomon (1968) adjustments for northern hardwoods in New Hampshire. Our growth trends also appear to

be consistent with the Carmean (1978) findings that red maple and beech maintain height growth better than other northern hardwood species. Furthermore, all of our projected site indices were consistent with those found by Leak (1978) in the White Mountains (Table 3).

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